Northern Wolves AC
55 9th St, 55-A2
Brooklyn, New York 11215
P: (347) 463-9248

Project: 25595 LA Fitness Eastchester Road
1776 Eastchester Road
Bronx, New York 10461

Submittal #1.0 - Trane RTU-12 Unit

Revision	0	**Submittal Manager**	Jonathan Maginnis (Northern Wolves AC)
Status	Open	**Date Created**	Oct 23, 2025
Issue Date	Oct 23, 2025	**Spec Section**	
Responsible Contractor	Northern Wolves AC	**Received From**	
Received Date		**Submit By**	
Final Due Date		**Lead Time**	
		Cost Code	15010 - SUBMITTALS
Location		**Type**	Product Data
Submittal Package			
Approvers			
Ball in Court	Jonathan Maginnis (Northern Wolves AC)		
Distribution			
Description	Trane RTU-12 Unit		

Submittal Workflow

Name	Sent Date	Due Date	Returned Date	Response	Attachments
General Information Attachments					LA Fitness Bronx NY Submittal 10-09-2025.pdf

Submittal

Prepared For:
All Bidders

Date: October 09, 2025

Job Name:
LA Fitness Bronx NY

Opportunity ID: 8328528

Trane U.S. Inc. is pleased to provide the following submittal for your review and approval.

Product Summary

Qty	Product
1	3 - 25 Ton Precedent Unitary Rooftop

Trey Clemmer, National Account Manager
Trane U.S. Inc.
2333 158th Court NE
Bellevue, WA 98008
E-mail: Trey.Clemmer@Trane.com
Office Phone: (425) 213-9111

The attached information describes the equipment we propose to furnish for this project and is submitted for your approval.

Submittal acceptance and return is a critical step, so please ensure submittals are returned with approval to release to production within 14 days of submittal date.

Product performance and submittal data is valid for a period of 6 months from the date of submittal generation. If six months or more has elapsed between submittal generation and equipment release, the product performance and submittal data will need to be verified. It is the customer's responsibility to obtain such verification.

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3 - 25 Ton Precedent Unitary Rooftop (Item A1) **3**

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Tag Data - 3 - 25 Ton Precedent Unitary Rooftop (Qty: 1)

Item	Tag(s)	Qty	Description	Model Number
A1	RTU-12	1	20 Ton PKGD Precedent Unitary Rooftop	YHK240A4SBL**M4E0A10100C001

Product Data - 3 - 25 Ton Precedent Unitary Rooftop
Item: A1 Qty: 1 Tag(s): RTU-12

DX Cooling / Gas Heat
High Efficiency
R-454B Refrigerant
20 Ton
460/60/3
Symbio 700 unit controls
Hail Guards
10:1 Modulating gas heat
Low capacity gas heat
0-100% Low Leak Economizer, ref. enthalpy control, with barometric relief damper
Multiple Zone Variable Air Volume with standard motor
Hinged access panels with 2-in MERV 13 filters
Through-the-base electric connections
Non-fused disconnect switch
Advanced Controller with BACnet communications interface
Return and Supply air smoke detector
Stainless steel drain pan

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Performance Data - 3 - 25 Ton Precedent Unitary Rooftop

Tags	RTU-12
Cooling Entering Dry Bulb (F)	80.00
Cooling Entering Wet Bulb (F)	67.00
Summer Ambient (F)	95.00
Heating Entering Air Temperature (F)	60.00
Design Airflow (cfm)	6000
Airflow Application	Downflow
Design ESP (in H2O)	1.300
Total SP (in H2O)	1.733
Elevation (ft)	0.00
Gross Total Capacity (MBh)	237.08
Gross Sensible Capacity (MBh)	162.60
Gross Latent Capacity (MBh)	74.48
Net Total Capacity (MBh)	228.63
Net Sensible Capacity (MBh)	154.16
Net Sensible Heat Ratio (%)	67.00
Coil LAT DB (F)	54.99
Coil LAT WB (F)	53.93
Cooling Leaving Unit Dry Bulb (F)	57.05
Cooling Leaving Unit WB (F)	54.79
Fan Motor Heat (MBh)	3.38
Dew Point Temperature (F)	53.21
Refrigerant charge - Ckt 1 (lb)	14.0
Saturated Discharge Temperature (F)	122.89
Saturated Suction Temperature (F)	47.98
Heat Static Pressure Adj (in H2O)	0.111
Component SP Add (in H2O)	0.322
Max Available ESP (in H2O)	1.567
Supply Motor Horsepower (hp)	3.000
Total Supply Operating Horsepower (hp)	3.180
Supply RPM (rpm)	1348
Compressor Power (kW)	17.73
System Power (kW)	22.07
EER @ AHRI	10.80
IEER @ AHRI	16.10
MCA (A)	55.00
MOP (A)	70.00
Compressor 1 RLA (A)	22.40
Compressor 2 RLA (A)	12.20
Condenser Fan FLA (A)	2.50
Evaporator Fan FLA (A)	4.60
Heating Input Capacity (MBh)	250.00
Output Heating Capacity (MBh)	202.50
Heating Leaving Air Temperature (F)	90.75
Heating Temperature Rise (F)	30.75
Height (ft)	4.92
Width (ft)	7.25
Length (ft)	10.25
Approx Installed Weight (lb)	2442.0
Corner weight A (lb)	754.0
Corner Weight B (lb)	516.0
Corner Weight C (lb)	337.0
Corner Weight D (lb)	493.0
Center of Gravity - Length (ft)	4.17
Center of Gravity - Width (ft)	2.83

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Tags	RTU-12
Ducted Discharge - 63 Hz (dB)	87
Ducted Discharge - 125 Hz (dB)	94
Ducted Discharge - 250 Hz (dB)	80
Ducted Discharge - 500 Hz (dB)	74
Ducted Discharge - 1 kHz (dB)	69
Ducted Discharge - 2 kHz (dB)	65
Ducted Discharge - 4 kHz (dB)	65
Ducted Discharge - 8 kHz (dB)	64
Ducted Inlet - 63 Hz (dB)	80
Ducted Inlet - 125 Hz (dB)	85
Ducted Inlet - 250 Hz (dB)	73
Ducted Inlet - 500 Hz (dB)	67
Ducted Inlet - 1 kHz (dB)	64
Ducted Inlet - 2 kHz (dB)	60
Ducted Inlet - 4 kHz (dB)	60
Ducted Inlet - 8 kHz (dB)	58
Outdoor Noise - 63 Hz (dB)	89
Outdoor Noise - 125 Hz (dB)	89
Outdoor Noise - 250 Hz (dB)	91
Outdoor Noise - 500 Hz (dB)	89
Outdoor Noise - 1 kHz (dB)	86
Outdoor Noise - 2 kHz (dB)	82
Outdoor Noise - 4 kHz (dB)	79
Outdoor Noise - 8 kHz (dB)	73
Acoustic Footnote 1	Ducted Discharge/Ducted Inlet prediction data conform to AHRI 260
Supply Fan Count	2
Indoor Fan Type	BC Plenum
Indoor Fan Drive Type	Variable Direct
Outdoor Fan Type	Propeller
Outdoor Fan Drive Type	Direct
Outdoor Fan Quantity	2
Heating Stages	10:1
Replication Run	297

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Mechanical Specifications - 3 - 25 Ton Precedent Unitary Rooftop
Item: A1 Qty: 1 Tag(s): RTU-12

General
Packaged rooftop unit cooling capacities, heating capacities, and efficiencies are certified to the following standards:
- 3 to 5 ton units: AHRI Standard 210/240.
- 6 to 25 ton units: AHRI Standard 340/360.
- Gas Heating Units: ANSI Z21.47 and 10 CFR Part 431 for Commercial Warm Air.
- Convertible airflow.
- Symbio? controls operating range between 40°F and 125°F in cooling mode standard from the factory. Field-installed low ambient kit extends operating range down to 0°F.
- Factory assembled, internally wired, fully charged, and 100 percent run tested to verify cooling operation, fan and blower rotation, and control sequence.
- Colored and numbered wiring internal to the unit for simplified identification.
- cULus listed and classified in accordance for Central Cooling Air Conditioners.
- Unit shall be furnished with a leak detection system from the factory

Casing
- Zinc coated, heavy gauge, galvanized steel.
- Weather resistant pre-painted metal with galvanized substrate.
- Meets ASTM B117, 672 hour salt spray test.
- Removable single side maintenance access panels.
- Lifting handles in maintenance access panels (can be removed and reinstalled by removing fasteners while providing a water and air tight seal).
- Exposed vertical panels and top covers in the indoor air section insulated with a cleanable foil-faced, fire-retardant permanent, odorless glass fiber material.
- Base pan shall have no penetrations within the perimeter of the curb other than the raised 1 inch high downflow supply/return openings to provide an added water integrity precaution, if the condensate drain backs up.
- Base of the unit insulated with 1/8 inch, foil-faced, closed-cell insulation.
- Unit base provisions for forklift and/or crane lifting on three sides of unit.

Hail Guards
- Provides condenser coil protection.

Stainless Steel Drain Pan
- Corrosion and oxidation resistance.
- Constructed of 304 stainless steel.

Microchannel Coils
- Optimal heat transfer performance due to flat, streamlined tubes with small ports, and metallurgical tube-to-fin bond.
- Reduce system refrigerant charge by up to 50% leading to better compressor reliability.
- Compact all-aluminum microchannel coils reduce the unit weight.
- Recyclable all aluminum coils All aluminium construction minimizes galvanic corrosion.
- Strong aluminum brazed structure provides better fin protection.
- Flat streamlined tubes more dust resistant and easy to clean.
- Coils leak tested at the factory to ensure the pressure integrity.

Compressors
- All units have direct-drive, hermetic, scroll type compressors with centrifugal type oil pumps.
- Suction gas-cooled motor with voltage utilization range of plus or minus 10 percent of unit nameplate voltage.
- Internal overloads standard with scroll compressors.
-Three stages of cooling available on 6 to 17.5 tons units and four stages of cooling available on 20 and 25 tons units.

Frostat
- Utilized as a safety device.
- Opens to prevent freezing temperatures on evaporator coil.

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- Temperature will need to rise to 50°F before closing.
- Utilized in low airflow or high outside air applications (cooling only).

Gas Heating Section

-The heating section shall have a progressive tubular heat exchanger with corrosion-resistant aluminized steel tubes and burners as standard on all models.
-Stainless steel heat exchanger with 409 stainless steel tubes and 439 stainless steel burners shall be optional.
- Induced draft combustion blower shall be used to pull the combustion products through the firing tubes.
- Heater shall use a direct spark ignition (DSI) system.
- On initial call for heat, the combustion blower shall purge the heat exchanger for 20 seconds before ignition.
- After three unsuccessful ignition attempts, entire heating system shall be locked out until manually reset at the thermostat/zone sensor.
- Units shall be suitable for use with natural gas or propane (field-installed kit).

Indoor Fan

- Direct drive plenum fan design - 6 to 25 tons units.
- Plenum fan design - backward-curved fan wheel along with an external rotor direct drive variable speed indoor motor.
- Supply fan speed adjustments can be made using the Symbio 700 or Mobile App.
- Motors are thermally protected.
- Variable speed direct drive motors are high efficiency - 6 to 25 tons.

Heat Exchanger

- Compact cabinet features a tubular heat exchanger in low, medium and high heat capacities.
- Corrosion-resistant aluminized steel tubes and burners are standard on all models.
- Induced draft blower to pull the gas mixture through the burner tubes.
- Direct spark ignition and a flame sensor as a safety device to validate the flame.

Through-the-Base Electrical with Disconnect Switch

- 3-pole, molded case, disconnect switch with provisions for through-the-base electrical connections.
- Disconnect switch installed within unit in a water tight enclosure.
- Wiring provided from the switch to the unit high voltage terminal block.
- Switch cULus agency recognized.

Note: *Disconnect switch sized per NEC and cULus guidelines but will not be used* in place of unit overcurrent protection.

Low Leak Economizer

- This accessory meets the low leak requirements for ASHRAE 90.1, IECC, and CA Title 24 standards (4 cfm/ ft2@1" wg exterior air/return air). This option allows for 100% outdoor air supply from 0% to 100% modulating dampers and comes standard with barometric relief. It can be paired with powered exhaust for additional building pressure relief. This option can be paired with or without fault detection and diagnostics (FDD) to meet the current mandatory CA Title 24 requirements. The economizers come with four control options: dry bulb, reference enthalpy, comparative enthalpy, or differential dry bulb.

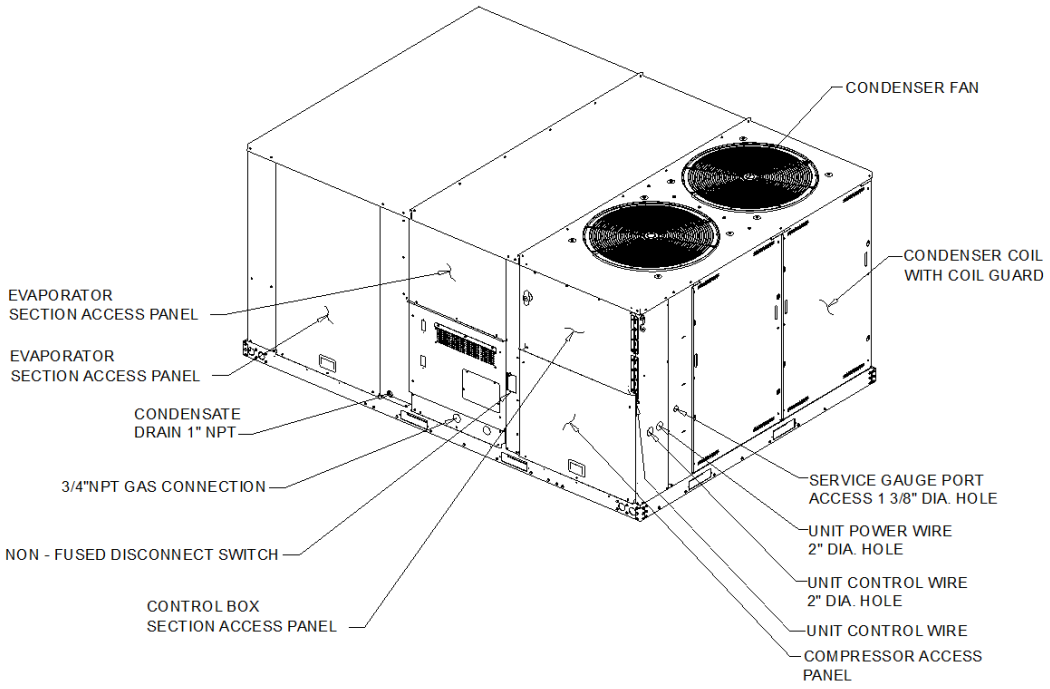
Reference or Comparative Enthalpy

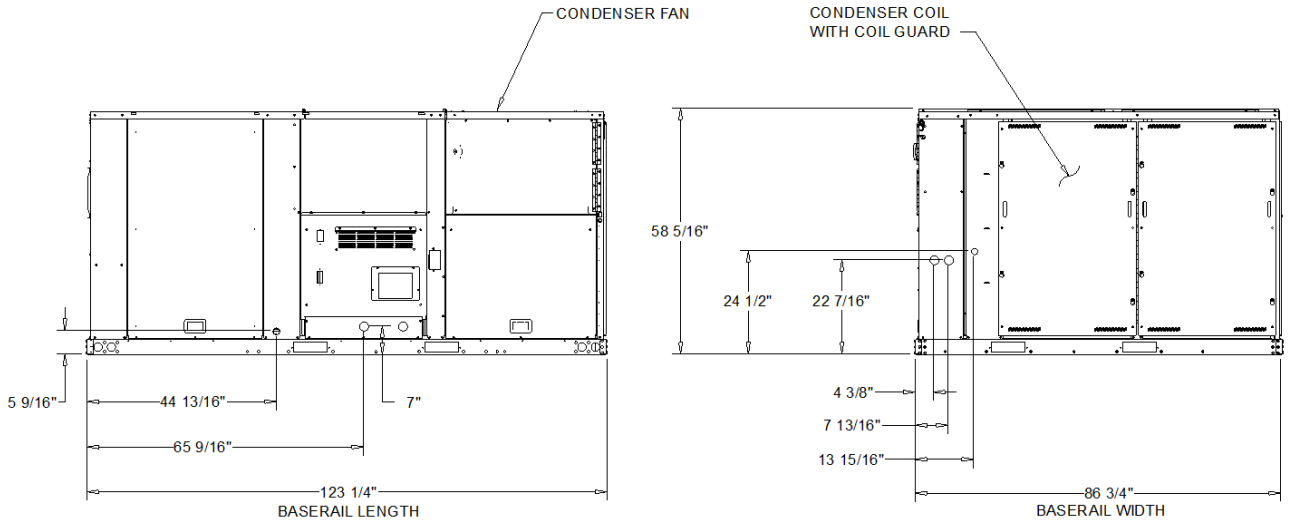
- Reference enthalpy used to measure and communicate outdoor humidity.
- Unit receives and uses information to provide improved comfort cooling while using the economizer.
- Comparative enthalpy measures and communicates humidity for both outdoor and return air conditions, and return air temperature.
- Unit receives and uses information to maximize use of economizer cooling, and to provide maximum occupant comfort control.
- Reference or comparative enthalpy available when a factory or field installed downflow economizer ordered.

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Dimensional Drawings - 3 - 25 Ton Precedent Unitary Rooftop
Item: A1 Qty: 1 Tag(s): RTU-12

NOTES:
1. THRU -THE -BASE ELECTRICAL IS NOT STANDARD ON ALL UNITS.
2. VERIFY WEIGHTS, CONNECTIONS, AND ALL DIMENSIONS WITH INSTALLER DOCUMENTS BEFORE INSTALLATION



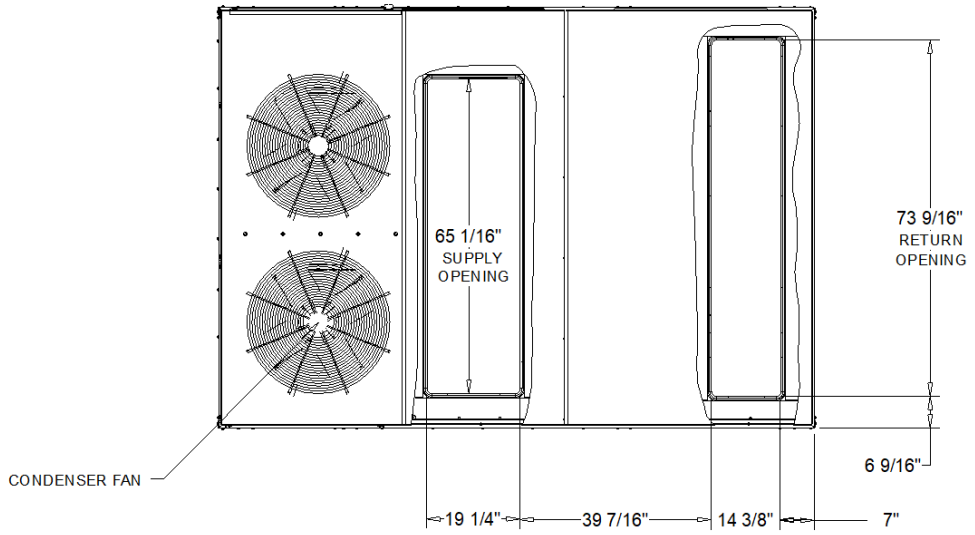


DX COOLING / GAS HEAT HIGH EFFICIENCY

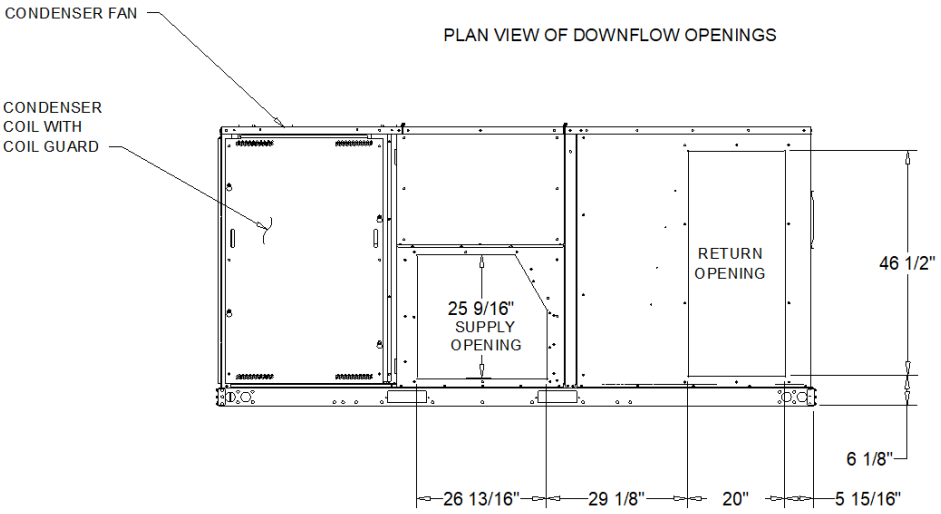
DIMENSION DRAWING

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Dimensional Drawings - 3 - 25 Ton Precedent Unitary Rooftop
Item: A1 Qty: 1 Tag(s): RTU-12



PLAN VIEW OF DOWNFLOW OPENINGS



HORIZONTAL AIR FLOW OPENING

DX COOLING / GAS HEAT HIGH EFFICIENCY

DIMENSION DRAWING

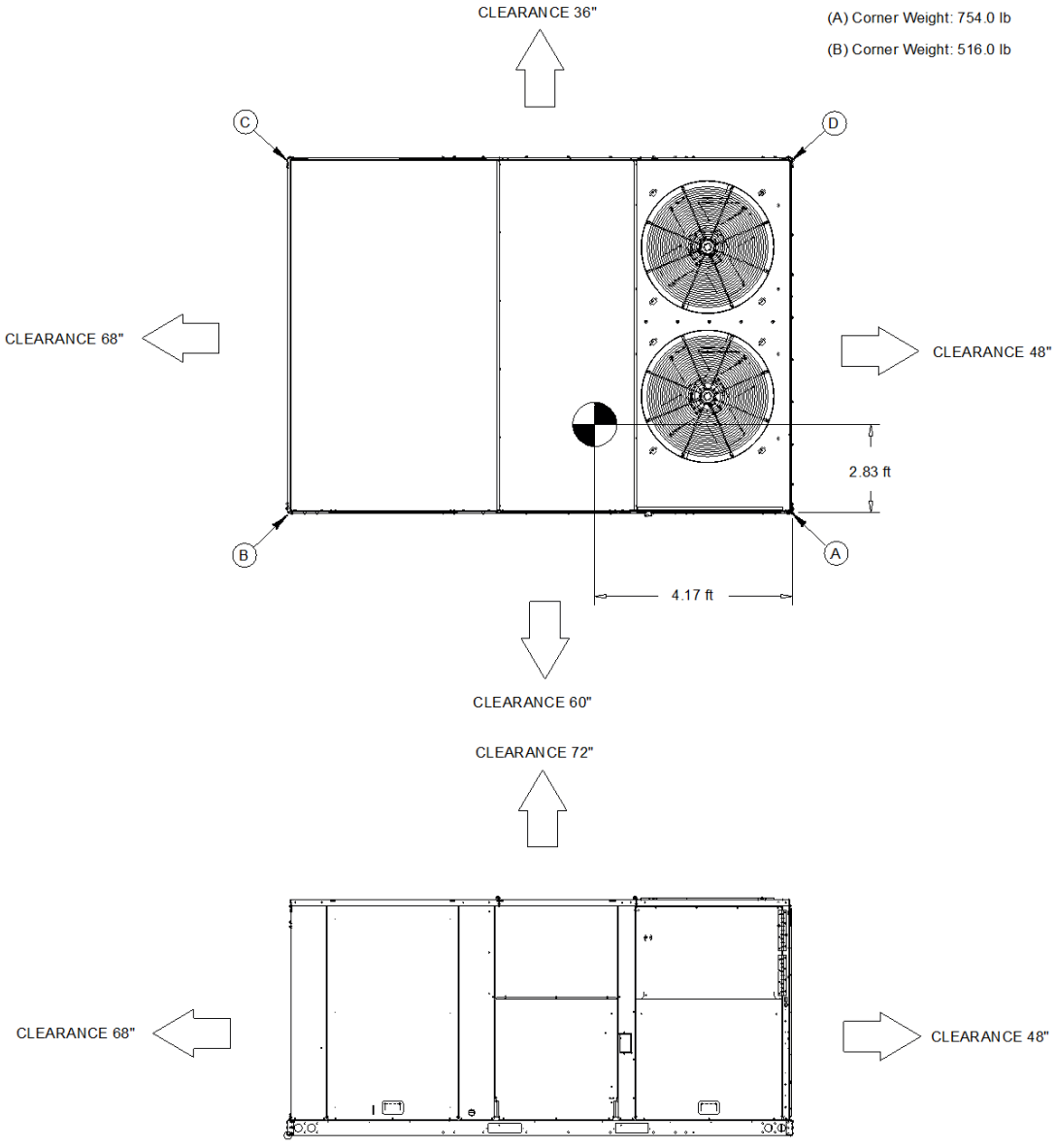
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Weight, Clearance & Rigging - 3 - 25 Ton Precedent Unitary Rooftop
Item: A1 Qty: 1 Tag(s): RTU-12

NOTES:
1. APPROX. INSTALLED WEIGHT INCLUDES ALL SELECTED OPTIONS AND ACCESSORIES.
2. CORNER WEIGHTS ARE FOR BASE UNIT ONLY AND DO NOT INCLUDE OPTIONS OR ACCESSORIES.
3. WEIGHT INCLUDES BOTH FACTORY AND FIELD INSTALLED ACCESSORY.

Approximate Installed Weight: 2,442.0 lb

(A) Corner Weight: 754.0 lb
(B) Corner Weight: 516.0 lb
(C) Corner Weight: 337.0 lb
(D) Corner Weight: 493.0 lb

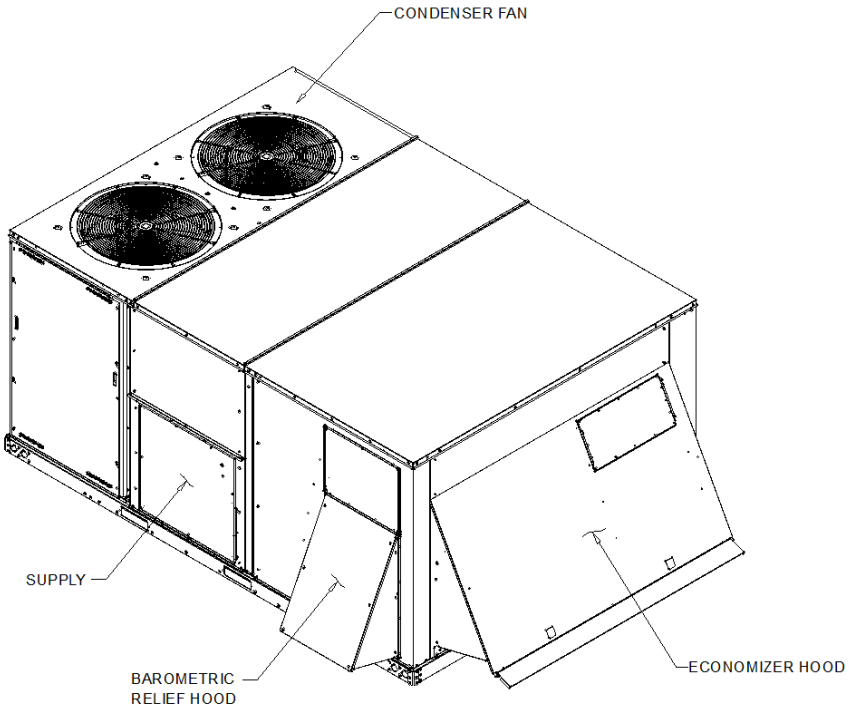


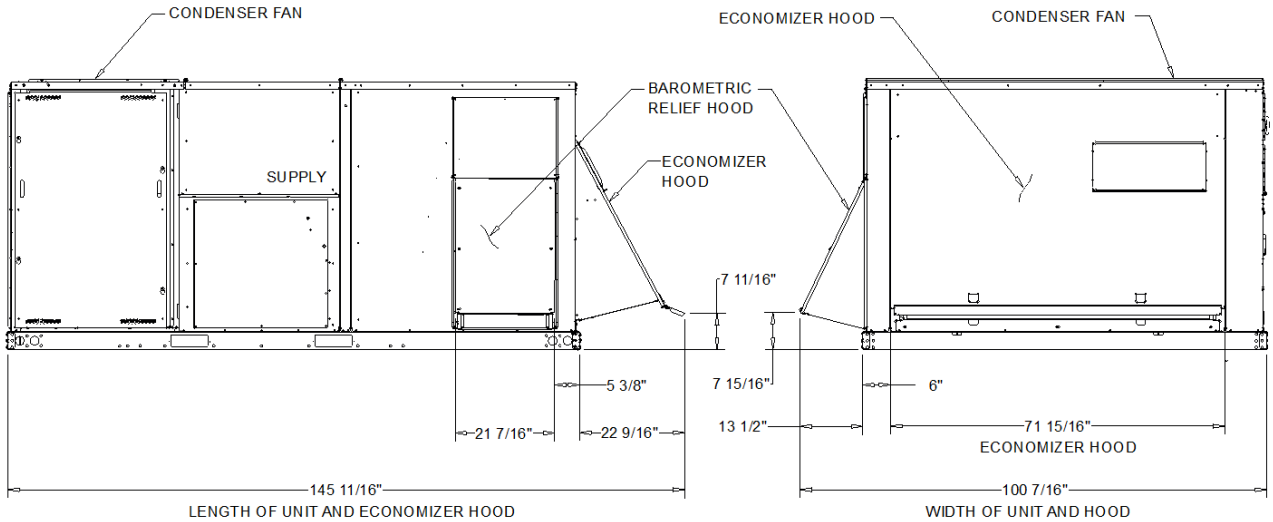
DX COOLING / GAS HEAT HIGH EFFICIENCY

WEIGHTS AND CLEARANCES

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Accessory - 3 - 25 Ton Precedent Unitary Rooftop
Item: A1 Qty: 1 Tag(s): RTU-12



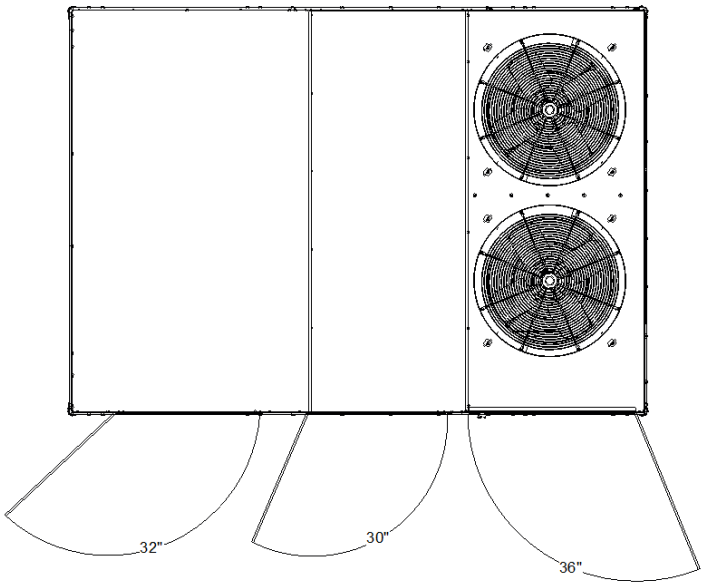


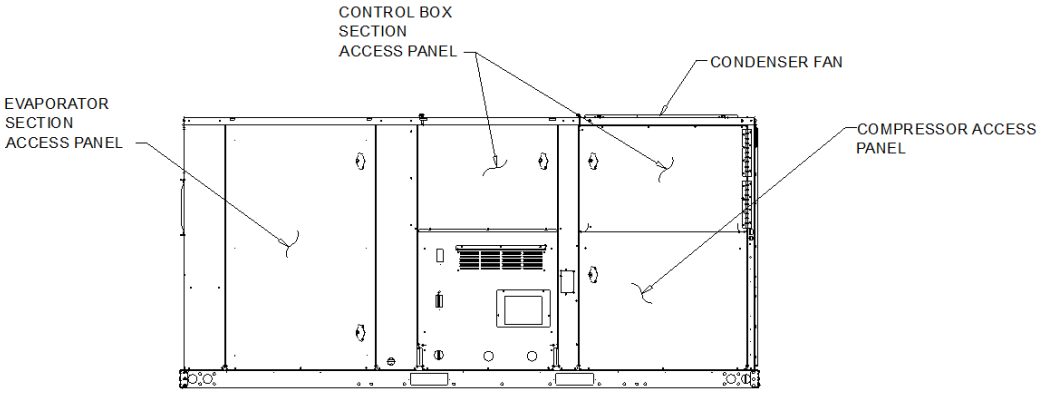
LOW LEAK ECONOMIZER AIR DAMPER

DX COOLING / GAS HEAT HIGH EFFICIENCY

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Accessory - 3 - 25 Ton Precedent Unitary Rooftop
Item: A1 Qty: 1 Tag(s): RTU-12



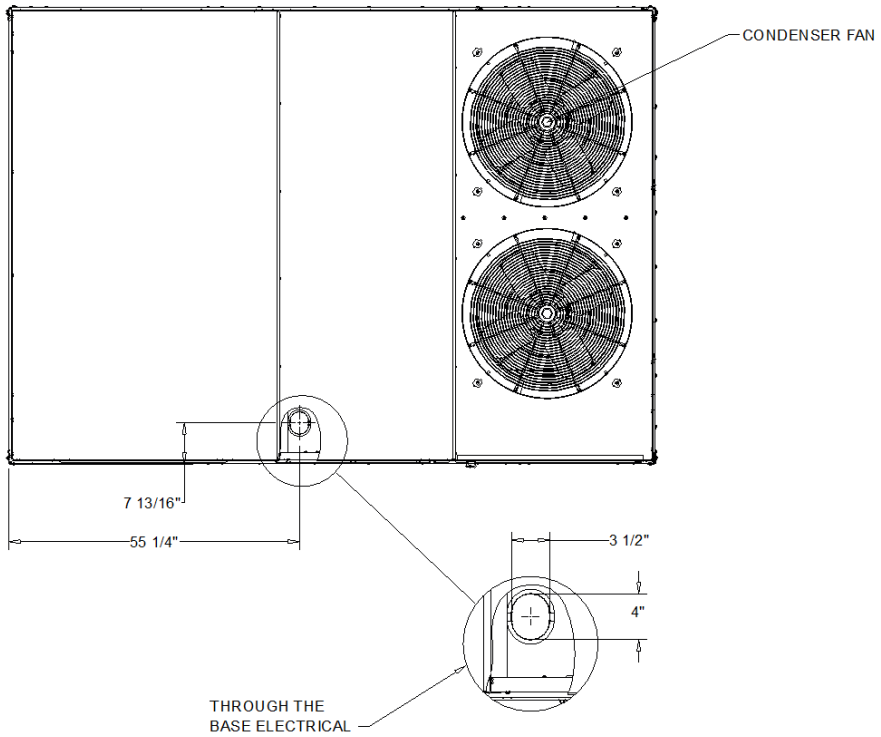


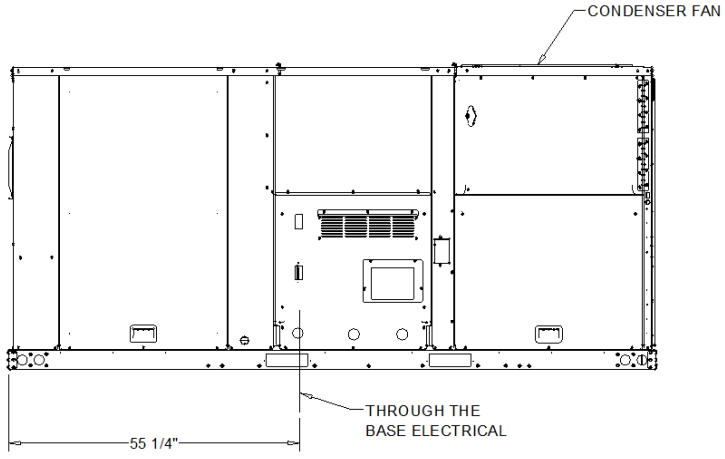
SWING DIAMETER FOR HINGED DOOR(S) (OPTION)

DX COOLING / GAS HEAT HIGH EFFICIENCY

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Accessory - 3 - 25 Ton Precedent Unitary Rooftop
Item: A1 Qty: 1 Tag(s): RTU-12





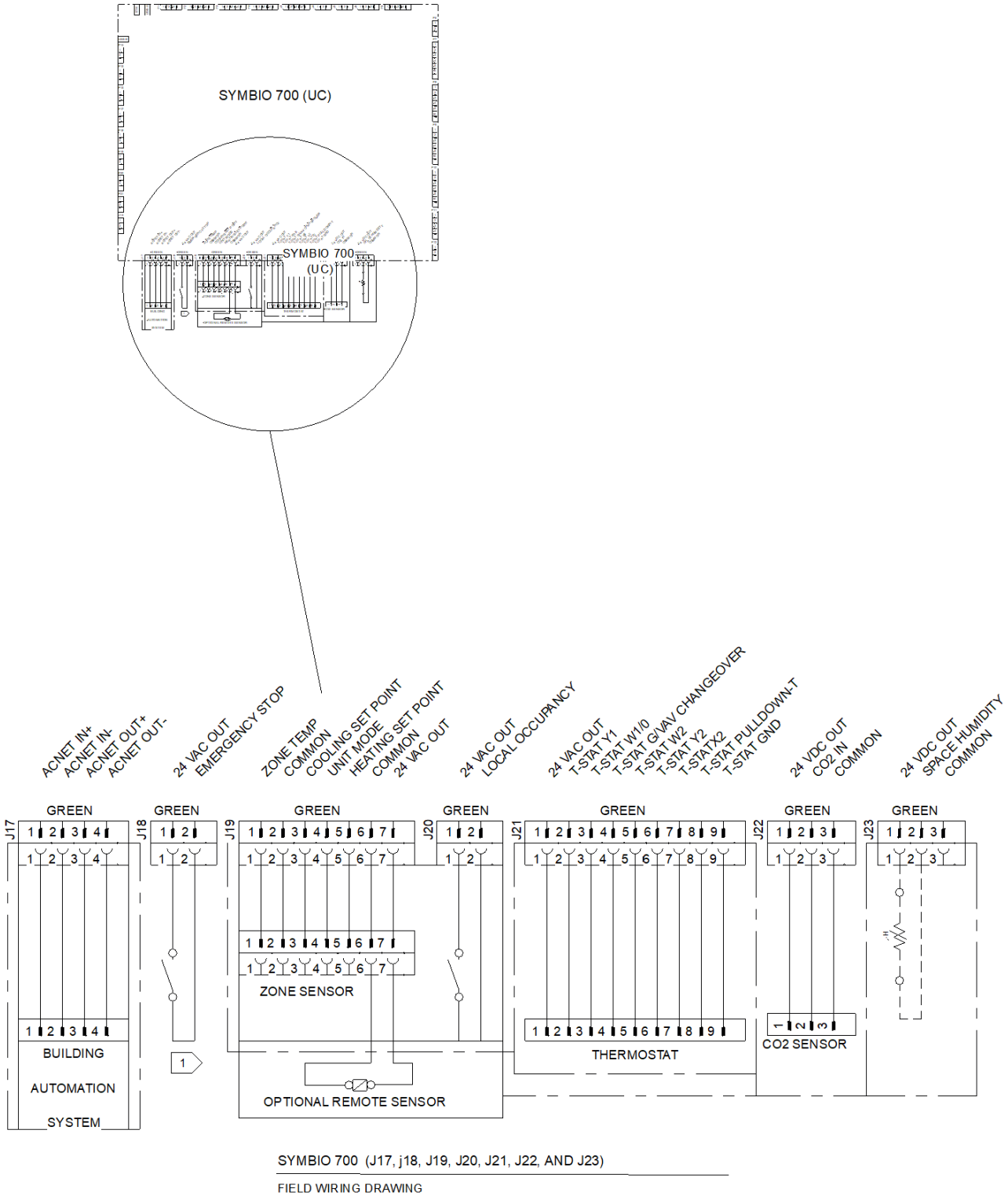
THROUGH-THE-BASE ELECTRICAL (OPTION)

DX COOLING / GAS HEAT HIGH EFFICIENCY

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Field Wiring - 3 - 25 Ton Precedent Unitary Rooftop
Item: A1 Qty: 1 Tag(s): RTU-12

NOTES:
1. VERIFY WEIGHT, CONNECTION, AND ALL DIMENSION WITH INSTALLER DOCUMENTS BEFORE INSTALLATION



SYMBIO 700 (J17, j18, J19, J20, J21, J22, AND J23)

FIELD WIRING DRAWING

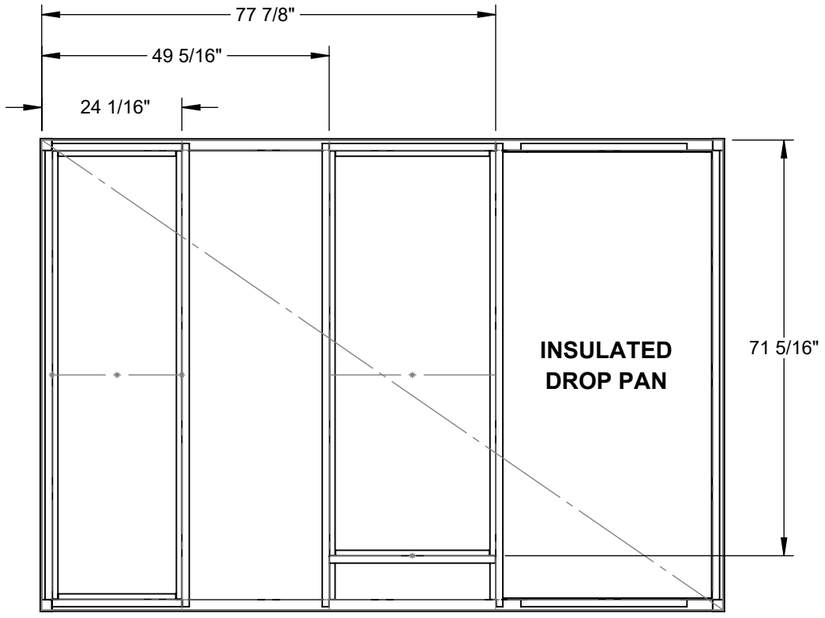
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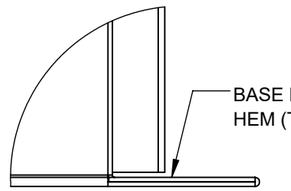
OPTIONS:
MATERIAL: GALVANIZED STEEL
1X4 WOOD NAILER (UNDER TOP FLANGE)
ANCHORING CURB TO ROOF FOR STEEL, CONCRETE OR WOOD, SEE DETAIL DRAWING "CURB ATTACHMENTS"

DIMENSIONED AND TOLERENCED PER ANSI Y14.5M-1982

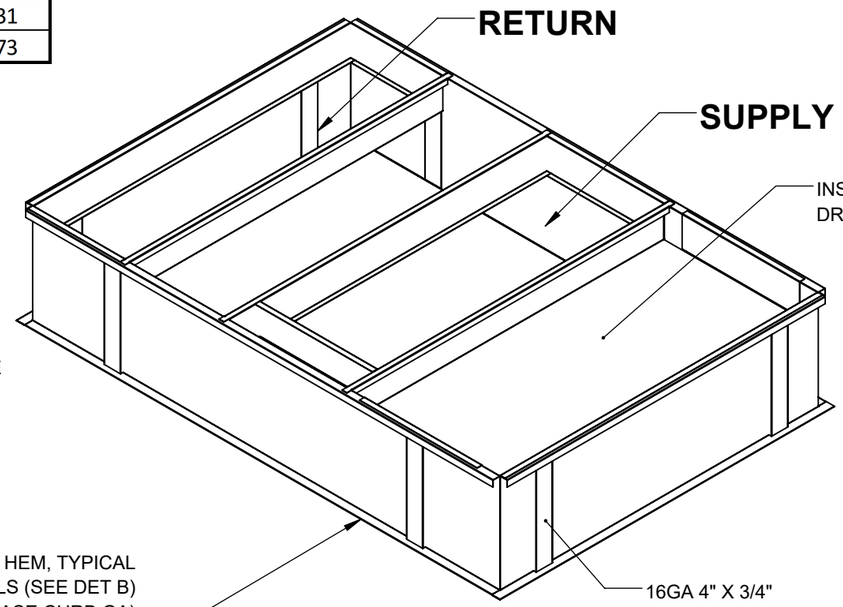
CDI P/N	CURB HT	MATERIAL GA	WEIGHT
C2-0011-3020	11	16	282
C2-0014-3020	14	16	303
C2-0018-3020	18	16	331
C2-0024-3020	24	16	373

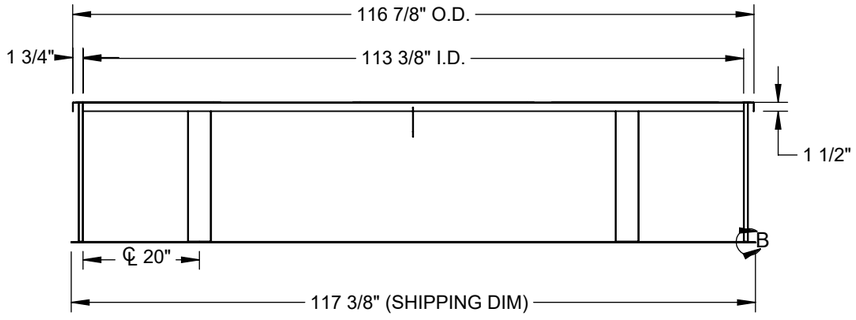
REVISIONS			
REV	DESCRIPTION	DATE	APPROVED
3	UPDATED IBC, FBC & CBC, REVISED MATL GA	3/1/21	SRP
4	ADDED WASHER PLATE NOTE	3/20/2021	ES
5	MADE 16 GA	4/21/22	KJ

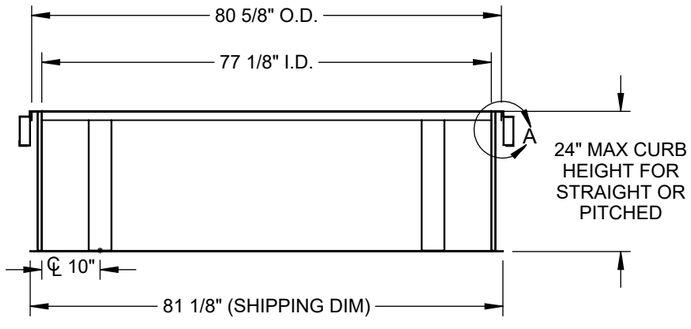


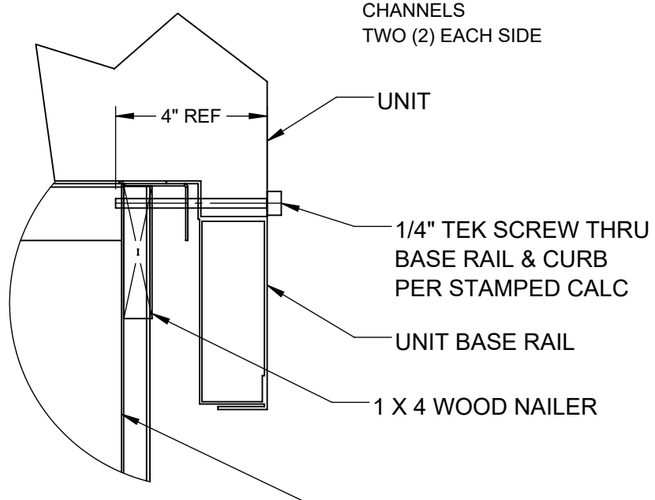


DETAIL B
SCALE 1 : 2









DETAIL A
SCALE 1 : 4

SEISMIC STRAIGHT CURB 3D ASSEMBLY

VAK Construction Engineering Services, LLC
8285 SW Nimbus Avenue, Suite 104
Beaverton, Oregon 97008
503.718.5999 tel 503.718.5980 fax
www.vakengineering.com

CURB DESIGNED TO MEET SEISMIC & WIND LOAD REQUIREMENTS FOR 2021 IBC/ASCE 7-22 & 2023 FBC FLORIDA BUILDING CODE
WIND DESIGN: MAX BUILDING HEIGHT 60FT; Kzt FACTOR = 1.00;
MAX 200 MPH ULTIMATE DESIGN WIND SPEED;
MAX 96 MPH ULTIMATE DESIGN TORNADO SPEED;
RISK CATEGORY II, III, OR IV BUILDING; EXPOSURE C
SEISMIC: Ss = 2.00 MAX, IMPORTANCE FACTOR Ip = 1.5, SDS = 1.33

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CURB SHOWN IS CDI STANDARD CONFIGURATION CDI RESERVES THE RIGHT TO CHANGE LAYOUT WITHOUT NOTIFICATION. IF CURB IS NEEDED IN A DIFFERENT CONFIGURATION CDI MUST BE NOTIFIED PRIOR TO PLACING AN ORDER.

CURB WEIGHT (LBS.): SEE TABLE

UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE:
.X ± 0.05
.XX ± 0.03
.XXX ± 0.01
Ø .XX= ± .007
<S ± 1/2°

DIMENSIONS ARE AS FOLLOWS: 1.00 in [25.40 mm]

3RD ANGLE PROJECTION

APPROVALS	DATE
DRAWN: kjones	
CHECKED: .	.

Tuesday, August 29, 2023 1:50:18 PM

CDI
17560 TYLER ST NW
ELK RIVER, MN 55330
(763)391-7790
(763)391-7851

TITLE: C2-00XX-3020

FILE NAME: C2-00XX-3020

ECO #:

DRAWING NO.:

REV: 5

SCALE: 1:26

SHEET: 4 OF 7

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